

Swastik Singh

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SKILLS

Languages:	Java, C++, Python, JavaScript, TypeScript, SQL
Core CS:	Data Structures & Algorithms, Object-Oriented Programming, DBMS
Frontend:	React, Next.js, HTML5, CSS3, Tailwind CSS
Backend:	Node.js, Express.js, Django REST Framework, Spring Boot, Celery, BullMQ, Socket.IO
Databases:	PostgreSQL, MongoDB, MySQL, Redis, Supabase
Tools / APIs:	Git, GitHub, Docker, Postman, Vitest, Google Gemini API
Concepts:	REST APIs, session/JWT auth, RBAC, multi-tenancy, Row Level Security, webhooks, idempotency, job queues

PROJECTS

VulnTrack | Vulnerability Management Platform (in development) | [GitHub](#) Sep 2026 – Present

- Built **session-based authentication** in Express.js and MongoDB: opaque 256-bit tokens stored only as SHA-256 hashes, Argon2id passwords, idle and absolute session expiry, Origin-header **CSRF checks**, and per-IP and per-account rate limits.
 - Designed **multi-tenant organizations with RBAC** (five roles, one central permission map); non-members get 404 for any organization, and a last-owner rule runs inside **MongoDB transactions** to stay correct under concurrent role changes.
 - Modeled an organization-scoped asset and software inventory that normalizes npm, PyPI, Maven and Go packages into **Package URL (purl)** identities, with optimistic concurrency; covered by 164 API tests.
- Tech Stack:** *Node.js, Express.js, MongoDB (Mongoose), React, Tailwind CSS, Zod, Vitest, Supertest, Docker*

Tenora | Multi-Tenant Property Billing SaaS | [GitHub](#) Sep 2026 – Present

- Built a **multi-tenant Django REST Framework** backend that resolves the tenant from the X-Tenant-ID header plus an active membership, never the request body; cross-tenant requests return 404 so other workspaces' records stay hidden.
 - Processed **Cashfree webhooks** behind a provider-neutral gateway adapter: raw-body signature checks, **idempotency via unique constraints**, and subscription changes only through a state-transition table under row locks.
 - Ran **Celery/Redis** sweeps for webhook retries, reminders, and reconciliation under **PostgreSQL advisory locks** so runs never overlap; deployed with Docker.
- Tech Stack:** *Python, Django REST Framework, PostgreSQL, Celery, Redis, Cashfree, React, TypeScript, Docker*

AI Interview Platform | Role-Based AI Hiring Platform | [GitHub](#) Aug 2026 – Present

- Built a four-role hiring platform (candidate, recruiter, interviewer, admin) as a TypeScript monorepo: **Express.js REST API** with Supabase Auth JWT, Zod validation, and rate limiting, plus role-guarded React routes.
 - Queued resume parsing on **BullMQ/Redis** with job IDs derived from the user and file hash to prevent duplicate processing; **Google Gemini** returns schema-constrained JSON (profile, ATS score, keyword gaps) validated with Zod.
 - Secured the Supabase PostgreSQL schema with **Row Level Security** and wrote a migration closing a **privilege-escalation** path that let any signed-in user set their own role to admin through the REST API.
- Tech Stack:** *TypeScript, Node.js, Express.js, React, Supabase (PostgreSQL, RLS), BullMQ, Redis, Google Gemini API, Vitest*

TRAINING

Data Structures & Algorithms | CipherSchools Jul 2025

- Completed a 70-hour program on arrays, stacks, queues, linked lists, and trees, with time and space complexity analysis.

CERTIFICATIONS

The Bits and Bytes of Computer Networking – Google	May 2026
Master Generative AI & Generative AI Tools	Apr 2026
Privacy and Security in Online Social Media – NPTEL (Elite)	Nov 2025
Fundamentals of Network Communication – University of Colorado	Oct 2024

EDUCATION

Lovely Professional University	Phagwara, Punjab
Bachelor of Technology in Computer Science and Engineering; CGPA: 7.1/10	Aug 2023 – Present
Nav Bharti Sr. Sec. School, New Delhi Intermediate (Class XII); Percentage: 75%	Mar 2022
Mahavir Senior Model School, New Delhi Matriculation (Class X); Percentage: 70%	Mar 2021